

YIFANG (MARK) CHEN

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EDUCATION

New York University Shanghai <i>B.S. in Computer Science; Minors in Mathematics and Data Science</i> Cumulative GPA: 4.0/4.0 GRE: 331 (Q170, V161) TOEFL iBT: 5.5/6.0 (110/120; Speaking: 5.5/6.0, Writing: 6.0/6.0)	Shanghai, China 2023.09 - 2027.05
- Awards: Baosteel Outstanding Student Scholarship (Top 2 across all majors and classes, ~0.25%); NYU Shanghai Recognition Award (Top 75, ~4%); Dean's List for Academic Year; Outstanding Student Worker of the Year - Major Courses: Machine Learning (A), Databases (A), Probability & Statistics (A), Multivariable Calculus (A)	
New York University <i>Study Away Student in Computer Science</i>	New York, USA 2025.09 - 2026.05
Major Courses: Natural Language Processing (A; Graduate Level), Numerical Analysis (A), Parallel Computing (A)	

RESEARCH EXPERIENCE

Research Assistant, NYU Shanghai <i>Unified Vision-Language Modeling for Heterogeneous Attributed Graphs</i> Advisor: Prof. Qiaoyu Tan Co-First Author, Manuscript Under Review	Shanghai, China 2025.03 - Present
- Developed a unified generative framework for reasoning across text-, image-, and multimodal-attributed graphs using a pretrained vision-language model as a shared backbone; designed target-aware textual retrieval and center-conditioned visual neighborhood adapters in the model's native token space - Outperformed all evaluated baselines across 9 in-domain and cross-graph transfer benchmarks, achieving 88.69% average in-domain accuracy and 66.67% average transfer accuracy, with improvements of 3.38 and 15.09 percentage points, respectively, and gains of up to 20.15 points on unseen graphs - Built end-to-end training, evaluation, and ablation pipelines on HPC clusters covering 9 datasets and 10 representative baselines; evaluated graph adapters, token budgets, neighborhood scaling, and optimization strategies, resulting in a co-first-authored manuscript under review at EMNLP 2026 (Avg. OA: 3.17)	

ACADEMIC PROJECTS

Language Model Fine-Tuning and Text-to-SQL Generation <i>Natural Language Processing Models for Structured Generation</i>	New York, USA 2026.04 - 2026.05
- Fine-tuned a T5-small encoder-decoder model for natural-language-to-SQL generation, implementing task-prefix formatting, SentencePiece tokenization, sequence truncation, dynamic batch padding, shifted decoder inputs for teacher forcing, and masked token-level cross-entropy loss - Developed structure-preserving augmentation that jointly replaced city names and airline entities in natural-language inputs and SQL targets, expanding the training set from 4,225 to 35,777 examples; improved held-out Record F1 by 7.54 points and achieved 79.86 Record F1 and 78.01 Record Exact Match - Built and optimized the end-to-end experimentation pipeline using AdamW with selective weight decay, cosine learning-rate scheduling with warmup, gradient clipping, and early stopping; implemented greedy and beam-search decoding, SQL execution-based evaluation, and systematic error analysis	
Pokémon Fusion Image Autoencoder <i>Deep Learning Model for Image Reconstruction and Classification</i>	Shanghai, China 2025.04 - 2025.05
- Developed a convolutional autoencoder for image classification, trained on 18,900 fusion Pokémon images, integrating ResBlock to compress images into an 8192-dimensional bottleneck for reconstruction and classification - Implemented data augmentation with random flips and rotations using torchvision, enhancing model robustness for classifying 170 Pokémon types, achieving high probing accuracy in linear probing tasks - Engineered a training pipeline with Adam optimization, early stopping, and ReduceLROnPlateau scheduling, achieving a test reconstruction MSE of 0.02 and classification loss of 0.15	

EXTRACURRICULAR EXPERIENCE

myGenius, Exam Preparation Platform Powered by Generative AI <i>Backend Developer</i>	Remote 2024.09 - 2024.12
- Developed a back-end system using Python and PDF-lib to preprocess user-uploaded PDFs into structured formats for AI-driven question generation, facilitating the creation of mock exam content from course materials - Engineered dynamic and customizable AI prompts for OpenAI's GPT-4 API, optimizing question relevance and alignment with user-selected difficulty levels to ensure the generation of relevant multiple-choice questions - Developed an optimized MongoDB system for efficient storage and retrieval of AI-generated content, facilitating seamless integration with a web interface for displaying multiple-choice questions under user-specific queries	
TAMID Group, NYU Shanghai <i>Principal of Tech Consultants</i>	Shanghai, China 2024.02 - 2024.12
- Led weekly project reviews with managers to track milestones and keep consulting engagements on schedule - Designed and coordinated a 15-session curriculum in web development and machine learning for 50+ participants	

SKILLS

- Python (PyTorch, Hugging Face, scikit-learn, NumPy), C, CUDA, MPI/OpenMP, Linux, Git, MySQL, LaTeX